



MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : PCC- CS302/PCC-CS302/PCCCS302 Computer Organisation

UPID : 003444

CS/B.TECH(N)/ODD/SEM-3/3444/2023-2024/1018

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (i) What are the different categories of memory/storage?
- (ii) Write one advantage of pipelining.
- (iii) Which type of program acts as an intermediary between a user of a computer and the computer hardware?
- (iv) The 2's complement of 15 is _____.
- (v) Name some types of devices used for Auxiliary Memory.
- (vi) What is interpreter ?
- (vii) What are the functions of the operating system?
- (viii) Carry, Overflow are also called _____.
- (ix) Which memory has the fastest speed in the computer memory hierarchy?
- (x) What is the full form of CISC?
- (xi) Which bus is bidirectional ?
- (xii) Which algorithms are based on add/subtract and shift category ?

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. What is Von Neumann bottleneck ? [5]
3. Describe IEEE 754 standard format for floating point representation. [5]
4. Explain the reading and writing operations of a SRAM [5]
5. Explain DMA controller. [5]
6. Explain the concept of hand shaking in IO operation. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. What are the advantages of Carry Look Ahead (CLA) over ripple carry adder? Explain with diagram. [7+8]
8. Explain the concept of virtual memory. What do you understand by page fault? [8+7]
9. Explain the difference between instruction pipeline and arithmetic pipeline. [15]
10. What is Operating System ? What are the different roles of an Operating System? [15]
11. Explain the difference between full associative and direct mapped cache mapping approaches . Explain "write through" and "write back" policies in cache. [8+7]

*** END OF PAPER ***